

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 31)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 1

. 1

Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO ₂ levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

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0 1

. 2

Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

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0 1

. 3

Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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0 1 . 4

Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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0 2

A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 2 . 1

Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 2 . 2

Explain how the reserve could be designed to provide a range of niches.

[3 marks]

0 2 . 3

Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

0 2 . 4

Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

10

0 3 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

Fuel

Food

Fibres

Medicines

0 3 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

0 3 . 3 Explain how forests regulate the hydrological cycle and local climate. [2 marks]

10

0 4 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 4 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture. [4 marks]

0 4 . 2

Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 4 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

[2 marks]

[5 marks]

05 . 3

Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

05 . 4

Describe the early stages of a lithosere.

[1 mark]

10

06 . 1

Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

06 . 2

Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

06 . 3

Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

06 . 4

Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

10

07

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

07 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer thousand tonnes

07 . 2 Calculate the maximum catch that would have kept the stock constant. **[1 mark]**

Answer thousand tonnes

07 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult. **[3 marks]**

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07 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing. **[4 marks]**

Catch per unit fishing effort

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Mean fish size and mean age

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0 8 . 1

Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest

☐

B Oceanic islands

☐

C Temperate broadleaf woodland

☐

D Mangroves

☐

0 8 . 2

Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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0 9 . 1 Explain how each of the following agricultural changes affects wildlife habitats. **[6 marks]**

Drainage of wet grassland

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Application of fertiliser to a species-rich meadow

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Removal of hedgerows to enlarge fields

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0 9 . 2 Explain why some farmland habitats are valuable for wildlife conservation. **[2 marks]**

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0 9 . 3 Describe how agriculture contributes to climate change. **[2 marks]**

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Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Do not write outside the box

Extra space

10 . **2** State what is meant by **industrial symbiosis**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS